Chirag Deepak Shinde

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Profile

Graduate Data Analyst experienced in delivering data-driven insights for academic and commercial stakeholders. Engineered automated ETL pipelines and Power BI dashboards for a Board Director Mortality & Stock Price Impact study, cutting manual data collection by 60% and accelerating stakeholder reporting. Built a sales forecasting platform at M.S Engineers, improving forecast accuracy by 15% and reducing reporting overhead by 30% through Snowflake data models and PySpark pipelines. Seeking to apply advanced analytics and data-engineering skills to drive strategic decision-making.

Skills

- **Analytics & Visualization**: Power BI (DAX, Semantic Modeling), Tableau (familiar), EDA, A/B Testing, Excel (Pivot Tables, VLOOKUP, Power Query), SSMS, Streamlit
- **Statistics & Modeling**: Statistical Modeling, Regression Analysis (OLS, Negative Binomial), ANOVA, Hypothesis Testing, Time-Series Analysis, Descriptive Statistics
- **Programming & Data**: Python (Pandas, NumPy), SQL, PySpark, PostgreSQL, CSV, JSON, Data Validation & Quality Engineering
- **Data Engineering & Cloud**: Azure Data Factory, Azure Synapse, Azure ADLS Gen2, Snowflake, Databricks, ETL/ELT, Data Modeling, Data Warehousing
- **Tools & DevOps**: Git, Docker, Power Automate, Dagster, CI/CD

Professional Experience

University of Illinois at Chicago — **Sep 2025 - May 2026**
Graduate Data Analyst — *Chicago, IL*
Project: Board Director Mortality & Stock Price Impact Study

- Reduced manual data collection effort by 60% by engineering automated ETL pipelines using Azure Data Factory and Gemini Flash LLM to extract and structure unstructured JSON and CSV web data into Azure Data Lake for analysis.
- Improved data quality for analysis by building PySpark validation and transformation workflows on Databricks, applying schema governance via Unity Catalog to deliver clean, analysis-ready datasets.
- Accelerated analytical reporting by profiling datasets, mapping entity relationships, and optimizing a normalized data model - cutting SQL query runtime and enabling seamless Power BI dashboard integration via Azure Synapse.
- Increased stakeholder outreach efficiency by 50% by automating alumni and startup campaign reporting pipelines using Power Automate integrated with Azure SQL.

M.S Engineers — **May 2023 - Jul 2024**
Data Analyst / BI Developer — *Pune, IND*
Project: Sales Forecasting & Analytics Data Platform

- Improved sales forecast accuracy by 15% by analyzing time-series data patterns and delivering clean, feature-engineered datasets from Azure Event Hubs ingestion pipelines to Python ML models - achieving 85% model accuracy.
- Enhanced BI reporting performance by designing Star Schema data models and structured data marts in Snowflake, supporting KPI dashboards and scalable reporting for business stakeholders.
- Ensured analytical data reliability by building a Spark SQL quality framework with null checks, schema drift detection, statistical threshold validation, and automated A/B testing on live data streams.
- Reduced manual reporting overhead by 30% by building medallion-style ETL pipelines with PySpark and Snowpark, standardizing data across Bronze, Silver, and Gold layers for consistent reporting.
- Streamlined analyst workflows by automating schema validation and model retraining via a CI/CD pipeline using Docker, GitHub Actions, and Git - reducing deployment overhead by 50%.

Projects

YouTube Mental Health Recovery Analysis — **Jan 2025 - May 2025**

- Extracted and structured 80+ features from YouTube API data using PySpark, Dagster, and RoBERTa/Gemini LLM workflows - enabling large-scale sentiment and engagement analysis.
- Uncovered statistically significant audience engagement drivers by applying Negative Binomial Regression, ANOVA, and OLS modeling to link emotional indicators with performance metrics.
- Reduced researcher report generation time by delivering interactive Power BI dashboards with optimized Python SQL queries for multi-dimensional, self-serve analysis.

RAG-based Q&A System with Vector DB — **Aug 2025 - Dec 2025**

- Built an end-to-end document ingestion and search pipeline using PySpark, FAISS vector indexing, and PostgreSQL metadata filtering - reducing manual document search time by 70%.
- Delivered a production-ready analytics application integrating a Gemini LLM backend with a Streamlit frontend for real-time, citation-grounded document querying.

Education

University of Illinois at Chicago (UIC) — **Aug 2024 - May 2026**
Master of Science, Computer Science — *Chicago, IL*

SPPU University — **Aug 2019 - May 2023**
Bachelor, Computer Engineering — *Pune, IND*

Certifications

- Databricks Certified Data Engineer Associate | Databricks | June 2026